Hackathon Problem Statement 1: Autonomous PCB Layout "Co-Pilot"

Develop an AI-driven middleware or agent that transforms high-level electronic system descriptions into validated, manufacture-ready hardware design files. The tool acts as a "Senior Hardware Engineer," automating component selection, pin-mapping, and netlist generation while ensuring electrical integrity and design rule compliance.

Core Requirements

- **Semantic Netlist Generation:** Extract system requirements (e.g., "LiPo powered ESP32 with motor drivers") and generate a structured Netlist defining all components and electrical connections.
- **Intelligent Component Selection:** Integrate with electronic distributor APIs (Octopart/Mouser/DigiKey) to select real-world part numbers based on stock availability, package size (SMD/THT), and electrical ratings.
- **Automated Pin-Mapping:** Resolve MCU pin assignments dynamically, ensuring no hardware peripheral conflicts (e.g., SPI vs. I2C) and optimizing for trace routing.
- **Electrical Safety Audit:** Perform automated "Pre-Flight" checks to flag risks like logic level mismatches (3.3V vs. 5V), missing decoupling capacitors, or insufficient power regulator current.

Technical Guidelines

- **LLM & RAG Integration:** Leverage LLMs (GPT-4o, Claude 3.5) with a RAG pipeline indexed with "Golden Circuit Snippets" and datasheet parameters for accurate circuitry.
- **EDA Automation:** Utilize Python APIs for popular EDA tools (e.g., **KiCad's pcbnew**) to programmatically generate schematic files (.kicad_sch) or layout scripts.
- **Knowledge Base:** Handle common sub-circuits: Power Management (LDO/Buck), USB-C PD, Battery Charging (TP4056/BQ series), and ESD Protection.
- **Verification:** Ensure output files are compatible with standard manufacturing workflows (Gerber generation or Web-EDA imports like EasyEDA/Flux).

Deliverables

- **Working Prototype:** A CLI tool, web dashboard, or plugin that processes natural language into design data.
- **Design Package:** For a given prompt, output a downloadable .zip containing the KiCad Schematic, a CSV Bill of Materials (BOM) with live links, and an Engineering Design Report.
- **Visual Proof:** A 2D schematic render or a generated "Connections Map" showing the logic of the circuit.
- **GitHub Repo:** Clean code with a setup script and documentation on API integrations used.